

Proposal Figure Checklist

A quick final-pass checklist for every figure in an observing proposal

1. Purpose & scientific argument

- ☐ Can I state in one sentence the conclusion I want the reviewer to draw?
- ☐ Does this figure advance the proposal narrative rather than just provide background?
- ☐ Is its job clear: impact, feasibility, or another necessary piece of the argument?
- ☐ If it is feasibility: does it show that the proposed observation or analysis will work?
- ☐ If it is impact: does it show what becomes possible or what changes if the proposal succeeds?
- ☐ If I removed this figure, would the proposal argument become meaningfully weaker?
- ☐ Was this figure designed for a proposal, rather than automatically recycled from a paper?

2. Communication & visual hierarchy

- ☐ Is the main scientific point the first thing my eye notices?
- ☐ Can I identify the key target, curve, regime, threshold, or result without hunting for it?
- ☐ Is supporting information present but appropriately de-emphasized?
- ☐ Would direct labels be clearer than making the reviewer repeatedly decode a legend?
- ☐ Are labels, ticks, annotations, and statistics large enough at final proposal size?
- ☐ Are line weight, style, marker size or shape, opacity, and color doing useful work?
- ☐ If panels should be compared, are their axes and scales normalized appropriately?
- ☐ Is the aspect ratio helping emphasize the signal or comparison I care about?
- ☐ Is there enough whitespace, or does the figure feel crowded?
- ☐ Thumbnail test: is the main point still apparent when the figure is small?

3. Color & accessibility

- ☐ Does color encode information rather than merely decorate the plot?
- ☐ Is the palette appropriate: categorical, sequential, or diverging?
- ☐ For a diverging scale, is the midpoint scientifically meaningful?
- ☐ Can important distinctions still be made in grayscale?
- ☐ Am I using redundant cues (line style, marker shape, labels) when a distinction matters?
- ☐ Are colors and styles used consistently across figures so the reviewer learns a visual vocabulary?

4. Polish & technical quality

- ☐ Does the figure feel deliberate rather than default?
- ☐ Is contrast strong enough without making every element dark, bright, or saturated?
- ☐ Are gridlines, spines, reference points, and secondary elements quieter than the result?
- ☐ Are typography, colors, and styling consistent with the rest of the proposal?
- ☐ Do shading, boxes, highlights, or other effects have a clear purpose?
- ☐ Am I using PDF/vector output where appropriate, or a genuinely high-resolution PNG?
- ☐ Have I viewed the figure at its actual size in the proposal PDF and checked another viewing mode?

5. Caption & proposal integration

- ☐ Does the figure make basic sense before I read the caption?
- ☐ Does the caption begin with an action-oriented takeaway rather than just describing the plot?
- ☐ Does the caption identify what is plotted without reproducing the main text?
- ☐ Does it explicitly connect the visual evidence to a benefit, requirement, or proposal claim?
- ☐ Have I stated the conclusion I want the reviewer to draw?
- ☐ Could a reviewer skimming only figures and captions recover most of the proposal argument?
- ☐ Does this figure connect naturally to the figures before and after it?

6. Final reviewer test

- ☐ 5 seconds: what do I notice first? Is it the intended thing?
- ☐ 15 seconds: can I tell why this figure matters to the proposal?
- ☐ Without the main text: can I understand the claim from the figure plus caption?
- ☐ With fresh eyes: is there anything I have to decode that I could simply label?
- ☐ Can I remove a panel, curve, annotation, color, or legend item without losing the argument?

Five questions to remember:

What conclusion? -> Do I see it immediately? -> Can I read it? -> Why does it matter? -> Does it earn the space?